PROBLEMS ON PRE-CALCULUS ALGEBRA MTH112

CHAPTER REVIEW

Exercise 1.

Let $A = \{x \in \mathbb{N} | x : 5\}$ and $B = \{x \in \mathbb{N} | x : 3\}$. Determine

a) $A \cap B$ b) $A \cup B$ c) $A \cup (A \cap B)$

Exercise 2. Let $A = [-2; 4]$, $B = (1; 5)$, $C = [-3; 3]$ be three sets of the universal set $X = [-5; 5]$ Determine

a) $A \cup B$ b) $A \cap (B \cup C)$ d) $\bar{A}$

e) $\bar{A} \cap (B \cup \bar{C})$ f) $\overline{A \cap B}$ g) $A \cup \overline{(B \cap C)}$

Exercise 4.

In Problems 9–20, use U = universal set = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}, A = {1, 3, 4, 5, 9}, B = {2, 4, 6, 7, 8}, and C = {1, 3, 4, 6} to find each set.

9. $A \cup B$	**10.** $A \cup C$	**11.** $A \cap B$	**12.** $A \cap C$
13. $(A \cup B) \cap C$	**14.** $(A \cap B) \cup C$	**15.** $\overline{A}$	**16.** $\overline{C}$
17. $\overline{A \cap B}$	**18.** $\overline{B \cup C}$	**19.** $\overline{A} \cup \overline{B}$	**20.** $\overline{B} \cap \overline{C}$

Exercise 5.

In Problems 65–68, determine the domain of the variable x in each expression.

65. $\frac{4}{x - 5}$ **66.** $\frac{-6}{x + 4}$ **67.** $\frac{x}{x + 4}$ **68.** $\frac{x - 2}{x - 6}$

Exercise 6.

In Problems 89–104, find the quotient and the remainder. Check your work by verifying that

(Quotient)(Divisor) + Remainder = Dividend

89. $4x^3 - 3x^2 + x + 1$ divided by $x + 2$

90. $3x^3 - x^2 + x - 2$ divided by $x + 2$

91. $4x^3 - 3x^2 + x + 1$ divided by x^2

92. $3x^3 - x^2 + x - 2$ divided by x^2

93. $5x^4 - 3x^2 + x + 1$ divided by $x^2 + 2$

94. $5x^4 - x^2 + x - 2$ divided by $x^2 + 2$

95. $4x^5 - 3x^2 + x + 1$ divided by $2x^3 - 1$

96. $3x^5 - x^2 + x - 2$ divided by $3x^3 - 1$

97. $2x^4 - 3x^3 + x + 1$ divided by $2x^2 + x + 1$

98. $3x^4 - x^3 + x - 2$ divided by $3x^2 + x + 1$

99. $-4x^3 + x^2 - 4$ divided by $x - 1$

100. $-3x^4 - 2x - 1$ divided by $x - 1$

101. $1 - x^2 + x^4$ divided by $x^2 + x + 1$

102. $1 - x^2 + x^4$ divided by $x^2 - x + 1$

103. $x^3 - a^3$ divided by $x - a$

104. $x^5 - a^5$ divided by $x - a$

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Exercise 7.

In Problems 75–122, factor completely each polynomial. If the polynomial cannot be factored, say it is prime.

75. $x^2 - 36$ **76.** $x^2 - 9$ **77.** $2 - 8x^2$ **78.** $3 - 27x^2$

79. $x^2 + 11x + 10$ **80.** $x^2 + 5x + 4$ **81.** $x^2 - 10x + 21$ **82.** $x^2 - 6x + 8$

83. $4x^2 - 8x + 32$ **84.** $3x^2 - 12x + 15$ **85.** $x^2 + 4x + 16$ **86.** $x^2 + 12x + 36$

87. $15 + 2x - x^2$ **88.** $14 + 6x - x^2$ **89.** $3x^2 - 12x - 36$ **90.** $x^3 + 8x^2 - 20x$

91. $y^4 + 11y^3 + 30y^2$ **92.** $3y^3 - 18y^2 - 48y$ **93.** $4x^2 + 12x + 9$ **94.** $9x^2 - 12x + 4$

95. $6x^2 + 8x + 2$ **96.** $8x^2 + 6x - 2$ **97.** $x^4 - 81$ **98.** $x^4 - 1$

99. $x^6 - 2x^3 + 1$ **100.** $x^6 + 2x^3 + 1$ **101.** $x^7 - x^5$ **102.** $x^8 - x^5$

103. $16x^2 + 24x + 9$ **104.** $9x^2 - 24x + 16$ **105.** $5 + 16x - 16x^2$ **106.** $5 + 11x - 16x^2$

107. $4y^2 - 16y + 15$ **108.** $9y^2 + 9y - 4$ **109.** $1 - 8x^2 - 9x^4$ **110.** $4 - 14x^2 - 8x^4$

111. $x(x + 3) - 6(x + 3)$ **112.** $5(3x - 7) + x(3x - 7)$ **113.** $(x + 2)^2 - 5(x + 2)$

114. $(x - 1)^2 - 2(x - 1)$ **115.** $(3x - 2)^3 - 27$ **116.** $(5x + 1)^3 - 1$

117. $3(x^2 + 10x + 25) - 4(x + 5)$ **118.** $7(x^2 - 6x + 9) + 5(x - 3)$ **119.** $x^3 + 2x^2 - x - 2$

120. $x^3 - 3x^2 - x + 3$ **121.** $x^4 - x^3 + x - 1$ **122.** $x^4 + x^3 + x + 1$

Exercise 8.

In Problems 123–132, expressions that occur in calculus are given. Factor completely each expression.

123. $2(3x + 4)^2 + (2x + 3) \cdot 2(3x + 4) \cdot 3$

124. $5(2x + 1)^2 + (5x - 6) \cdot 2(2x + 1) \cdot 2$

125. $2x(2x + 5) + x^2 \cdot 2$

126. $3x^2(8x - 3) + x^3 \cdot 8$

127. $2(x + 3)(x - 2)^3 + (x + 3)^2 \cdot 3(x - 2)^2$

128. $4(x + 5)^3(x - 1)^2 + (x + 5)^4 \cdot 2(x - 1)$

129. $(4x - 3)^2 + x \cdot 2(4x - 3) \cdot 4$

130. $3x^2(3x + 4)^2 + x^3 \cdot 2(3x + 4) \cdot 3$

131. $2(3x - 5) \cdot 3(2x + 1)^3 + (3x - 5)^2 \cdot 3(2x + 1)^2 \cdot 2$

132. $3(4x + 5)^2 \cdot 4(5x + 1)^2 + (4x + 5)^3 \cdot 2(5x + 1) \cdot 5$

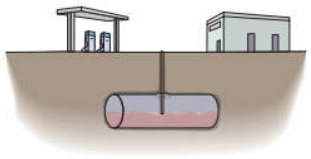
133. Show that $x^2 + 4$ is prime.

134. Show that $x^2 + x + 1$ is prime.

Exercise 9.

107. Calculating the Amount of Gasoline in a Tank A Shell station stores its gasoline in underground tanks that are right circular cylinders lying on their sides. See the illustration. The volume V of gasoline in the tank (in gallons) is given by the formula

$$V = 40h^2\sqrt{\frac{96}{h} - 0.608}$$



where h is the height of the gasoline (in inches) as measured on a depth stick.
(a) If $h = 12$ inches, how many gallons of gasoline are in the tank?
(b) If $h = 1$ inch, how many gallons of gasoline are in the tank?

HOMEWORKS:
Exercise 4: 17-20,
Exercise 7: 100-102
Exercise 9